

HENRY FREED

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PROJECTS

Winnow

Pruning MoEs with Variable Sized Experts

- Extended REAP from whole-expert removal to channel-level pruning within each expert, scoring channels by router weight $\times$ activation magnitude $\times$ down-projection column norm; prunes experts to variable widths instead of dropping them.
- Beats REAP on OLMoE across keep ratios; competitive on Qwen3.6-35B-A3B. Healed low-keep-ratio models with on-policy logit distillation.

Variable Sized MoEs

Mixed-Width Experts Within One MoE

- Designed and managed end-to-end research program studying compute allocation in MoE language models, from hypothesis development through evaluation methodology to published findings.
- Built evaluation framework measuring token routing patterns across domains; discovered routing correlates with contextual ambiguity, not computational difficulty. Code tokens route 14σ below cross-domain mean.
- Validated reproducibility across multiple random seeds, identifying that 30% of routing decisions show stable learned structure.
- Extended nanoGPT with MegaBlocks sparse kernels to support variable-sized expert configurations at 125M–250M parameter scale.

Variable FlexOlmo

Variable-Width Expert Pruning for FlexOlmo

- Pruned Ai2's FlexOlmo math expert to three widths (74%, 50%, 19%) using importance-based neuron scoring and KLD distillation on 228M tokens. The 74%-width expert matches baseline; the 19%-width expert lifts GSM8K from 13% to 44%.
- Designed evaluation comparing domain-specific vs. general importance analysis for pruning; found 58% neuron disagreement between methods, with domain-calibrated pruning producing measurably better training dynamics.
- Released logprobs dataset and trained model weights on HuggingFace.

Open Concept Steering

Mechanistic Interpretability

- Developed and open-sourced a method for extracting interpretable concepts from language models using sparse autoencoders (SAEs), training a 65k-dimensional SAE on 600M activations from OLMo-2-7B.
- Released full reproducibility package: interactive demo, blog post, dataset, training code, and model weights.

EMPLOYMENT

Remitly

Knowledge Development Engineer (Contract)

April 2026 - September 2026

Seattle, WA

- Build agentic AI / LLM automation for compliance analytics workflows, translating ambiguous domain requirements from a non-technical team into evaluable systems.
- Reverse-engineered and documented undocumented data provenance across orphaned analytics pipelines to make compliance reporting auditable.

Cirtec Medical, Volant Partners
AI/ML Engineer - Generative AI (Contract)

June 2024 - September 2025
Remote

- Owned full project lifecycle as sole AI technical resource: gathered requirements from marketing leadership, scoped solutions, built and deployed systems, and defined quality metrics—all on 1–2 week delivery cycles.
- **Cirtec Chat** – Designed RAG-based knowledge system end-to-end. Implemented retrieval pipeline, built evaluation framework (95% recall), and created usage analytics dashboard to track quality and inform iteration.
- **Content Pipeline** – Architected cross-platform automation system, aligning marketing, brand, and technical requirements into a consistent content delivery pipeline.
- Translated qualitative stakeholder feedback into technical requirements and measurable success criteria across all projects.

VanEck Securities
Emerging Markets Fixed Income Intern

June - September 2022
New York, NY

- Built Python tooling to automate sales presentation generation, reducing task completion time from 30 minutes to 3 minutes.

Portland Pickles Baseball Team
Assistant Director of Operations

July - August 2021
Portland, OR

- Implemented Kanban system for pre-game operations; managed interns and oversaw game-day logistics.
- Collected and tracked player performance metrics (exit velocity) using Trackman; managed live-stream production.

Naka and Adana Restaurants
Demi Sous Chef

February 2016 - November 2017
Seattle, WA

- Led kitchen operations under chef Shota Nakajima; managed transition from Naka to Adana including developing and deploying an entirely new dessert program on a compressed timeline.

TECHNICAL SKILLS

Languages & Tools	Python (PyTorch, HuggingFace Transformers, vLLM, Prime Intellect Verifiers), $\text{\LaTeX}$, Git
LLM & ML	Training, Distillation (incl. On-Policy), LoRA, RL, Inference, Evaluation Pipelines, MoE Architectures, Sparse Autoencoders, RAG Systems

EDUCATION

Portland State University <i>B.S. Applied Mathematics</i>	Graduated August 2023 <i>Portland, OR</i>
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INTERESTS

Homebrewing beer & cider, baseball analytics, the Grateful Dead, pawpaw & quince fruits